

Cloud-Robust Field-Scale Crop Intelligence with Sentinel-1/Sentinel-2 Fusion: An AgriSense Study Linking Karnal Operations, Haridwar Phenology and AgriFieldNet Crop Labels

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Abstract - AgriSense combines seasonal crop monitoring with two remote-sensing research workflows in one application. The Karnal component aligns ten satellite dates from 4,311 sample locations with a nineteen-period weather calendar and serves prototype crop, phenology and stress models. Its derived targets are kept separate from externally sourced supervision. The Haridwar component uses wheat-stage dates published by Kumar and Khare for a five-class secondary analysis of 59 acquisition rows across two seasons. Its selected fusion Random Forest reaches leave-one-season-out macro-F1 0.493. The completed AgriFieldNet extension uses the crop labels credited to Radiant Earth Foundation and IDinsight, adds 21 ten-day Sentinel-1/Sentinel-2 feature bins to 5,551 training fields, and compares 80 classical configurations and three deep sequence models under four spatially grouped folds. The fusion HistGradientBoosting leader obtains 73.30% accuracy and macro-F1 0.391, close to 0.389 for the temporal optical leader. Under 60% random optical masking, their mean macro-F1 values are 0.259 and 0.194. These results support the resilience of the selected fusion pipeline under the tested missing-signal scenario, while rare-class failures and model-selection effects limit broader claims. The contribution is the additional processing, explicit validation boundaries and application integration built around credited data sources.

Index Terms - Sentinel-1, Sentinel-2, crop classification, phenology, optical gaps, spatial validation, AgriFieldNet, AgriSense.

I. INTRODUCTION

Crop monitoring depends on a sequence of observations, not simply a clear image on one date. Optical indices describe vegetation condition, but their availability can change sharply during a cloudy season. Radar supplies complementary measurements, and a practical application must explain when either source is absent or old. Sentinel-1, Sentinel-2 and Google Earth Engine provide the observation and processing foundation used here [1-3].

Previous work has already established multisensor crop mapping. Rao et al. examined smallholder crop classification using Sentinel-1, Sentinel-2 and Planet imagery [4], while Tufail et al. studied combined SAR and optical time series [5]. Wozniak et al. investigated temporal radar descriptors [6]. AgriSense builds on this literature; it does not claim the first radar-optical classifier.

The same care is needed for supervision. Field-supported wheat phenology has been studied previously [7]. Kumar and Khare published the Haridwar observations used as stage anchors in the present secondary analysis [8,9]. AgriFieldNet is an established dataset with surveyed crop labels and prior challenge solutions [10,11]. Our work adds processing and evaluation around these sources and connects the outputs to the AgriSense application [12].

The application has three linked components with different purposes: a Karnal operational demonstration, a Haridwar phenology research page and an AgriFieldNet crop benchmark. Reporting their evidence separately is central to interpreting the platform correctly.

II. STUDY POSITIONING AND CONTRIBUTIONS

The study asks whether one application can make multisensor agricultural outputs useful without hiding differences in label origin, observation coverage and validation design. The contribution has four parts: an explicit field-time representation; a credited Haridwar secondary analysis; a completed temporal AgriFieldNet benchmark; and interfaces that expose the resulting evidence. The research components are linked to, rather than substituted for, the Karnal advisory demonstration.

III. DATA SOURCES, PROVENANCE AND ATTRIBUTION

Component	Original evidence	AgriSense contribution
Karnal	Copernicus satellite data [1,2], processed in Earth Engine [3]; derived application targets.	Observation alignment, coverage audit, inference service and prototype water rules.
Haridwar	Published wheat-stage chronology of Kumar and Khare [8,9].	Five-stage grouping, satellite-feature reanalysis, LOSO evaluation and robustness tests.
AgriFieldNet	Radiant Earth Foundation and IDinsight crop labels [10].	Field reconstruction, temporal feature extension, spatial benchmark and research dashboard.

TABLE I. Source ownership and the role of the present analysis.

A. Karnal Operational Layer

Karnal contains 4,311 fixed sample locations over ten satellite dates from June to October 2025, giving 43,110 records. A separate nineteen-step, eight-day weather calendar controls the application. The latest valid optical and radar observations are retrieved independently on or before each master date, and their source date and age are retained. These are sample locations, not a verified cadastral inventory. Crop, stress and phenology targets are derived, so the component supports engineering and coverage claims rather than independent agronomic accuracy.

B. Haridwar Phenology Evidence

The source chronology contains ten published wheat-stage anchors in each of the 2023-24 and 2024-25 seasons [8,9]. AgriSense groups them into Establishment, Vegetative, Reproductive, Grain Filling and Senescence/Harvest. Midpoint-based intervals assign labels to satellite acquisitions. The resulting table has 59 labelled rows within a 68-row calendar, with class supports of 12, 23, 16, seven and one. The grouping and acquisition labels are derived products, not direct observations at every acquisition.

C. AgriFieldNet Crop Labels

The original challenge contains 7,081 fields, including 5,551 training and 1,530 test fields, and thirteen crop/fallow classes [10]. The present benchmark evaluates only the 5,551 labelled training fields through cross-validation. The saved temporal batches cover 1 November 2021 to 29 May 2022 in 21 ten-day bins, yielding 116,571 field-period rows. These temporal features extend the original challenge input. The labels support crop-type classification; they do not supply field-date phenology or irrigation supervision.

IV. METHODOLOGY

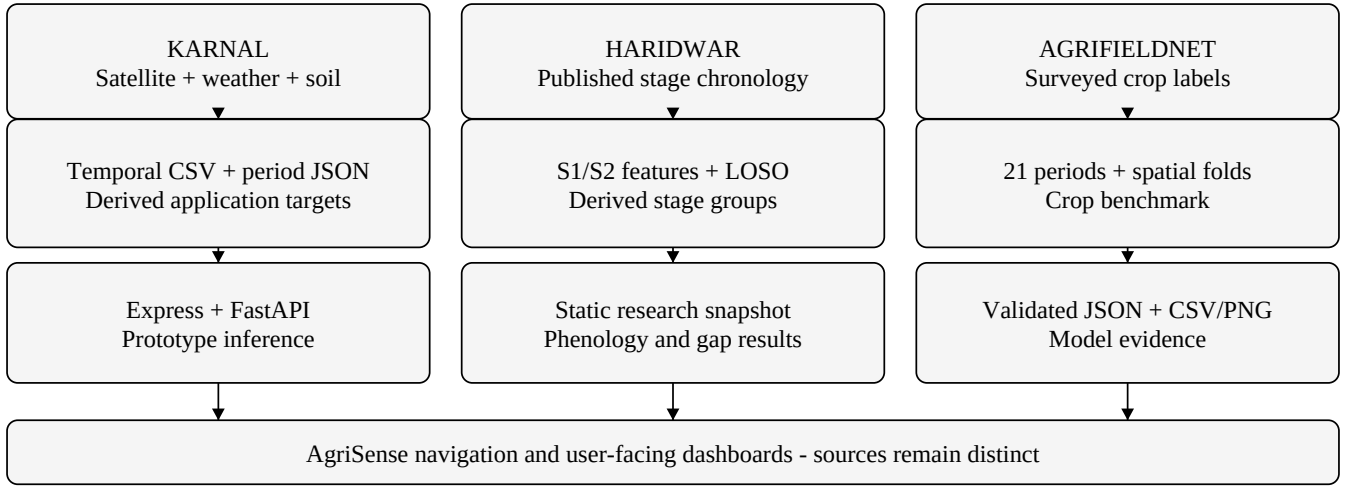


Figure 1. Three application components retain separate data and validation paths.

A. Common Field-Time Representation

The pipelines store optical signal features, radar backscatter and quality information together while preserving missingness. Karnal uses vegetation and moisture indices with VV, VH and their dB difference. The AgriFieldNet extension includes Sentinel-2 bands and indices, clear-observation diagnostics, and separate ascending and descending Sentinel-1 summaries. Values reduced to a finer grid do not acquire finer native spatial information.

B. Haridwar Secondary Classification and Validation

S2-only, S1-only and fusion feature families are evaluated with leave-one-season-out testing. A full season is withheld in each fold, and the two held-out prediction sets are pooled. Macro-F1 is the primary measure because stage support is uneven. The selected leaders are chosen using these same results. Consequently, their reported scores are comparative LOSO estimates after selection rather than performance on a separate final test. Transition estimates from smoothed NDVI are evaluated separately in days.

C. AgriFieldNet Temporal Extension and Spatial Validation

Field reconstruction preserves global identifiers and crop labels, and the completed run includes all eight periodic-feature and eight observation-template batches. Five feature families compare static S2, temporal S2, temporal S1, temporal fusion and compact temporal descriptors. Sixteen classical estimators per family produce eighty configurations; CNN1D, GRU and LSTM add three sequence configurations.

Representative field points define 0.10-degree spatial grid groups. Four stratified group folds keep each group in one partition, and every training fold must contain all thirteen classes. Median imputation with missing-value indicators and scaling are fitted within the training partition. Pooled OOF metrics use one held-out prediction per field per configuration. This design reduces local spatial overlap but does not establish transfer to unseen regions or seasons.

D. Optical-Gap Tests and Importance

For AgriFieldNet, selected leaders are fitted once per fold and reused for held-out masking tests. S2 signal columns are removed from approximately 20%, 40% or 60% of periods, or from two or four consecutive bins. Radar and S2 quality/count diagnostics remain. Each non-baseline scenario has fifteen repeats; the baseline has one. Repeat SD measures mask sensitivity, not generalization uncertainty. Saved permutation importance is exploratory: two repeats and at most 400 held-out fields per fold, with signed changes in macro-F1 retained.

V. RESULTS

A. Haridwar LOSO Secondary Analysis

Feature family	Selected model	Accuracy	Macro-F1
S2 only	RBF SVM	0.559	0.461
S1 only	Random Forest	0.644	0.429
S1 + S2	Random Forest	0.644	0.493

TABLE II. Pooled LOSO results for the derived Haridwar stage groups.

The fusion model attains balanced accuracy 0.504 and has the highest macro-F1 among the selected family leaders. S1-only matches its overall accuracy but not its class-balanced performance. These are AgriSense secondary-analysis results, not metrics reported by the original field-study authors. The one-row Senescence/Harvest class sharply limits what can be inferred about terminal-stage recognition.

B. AgriFieldNet Spatial Classification

Feature family	Leader	Accuracy	Macro-F1
A0 static S2	CatBoost	0.577	0.260
A1 temporal S2	CatBoost	0.695	0.389
A2 temporal S1	CatBoost	0.689	0.340
A3 temporal fusion	HistGradientBoosting	0.733	0.391
A4 descriptors	CatBoost	0.707	0.371

TABLE III. Selected family leaders from four spatially grouped folds.

The fusion leader obtains macro-F1 0.390976, balanced accuracy 0.379838, weighted F1 0.717539, kappa 0.634624 and MCC 0.635706. Its macro-F1 exceeds the temporal optical leader by only 0.002342. The selected fusion estimator also differs from the optical estimator, so this comparison does not isolate a causal sensor effect or demonstrate statistical superiority.

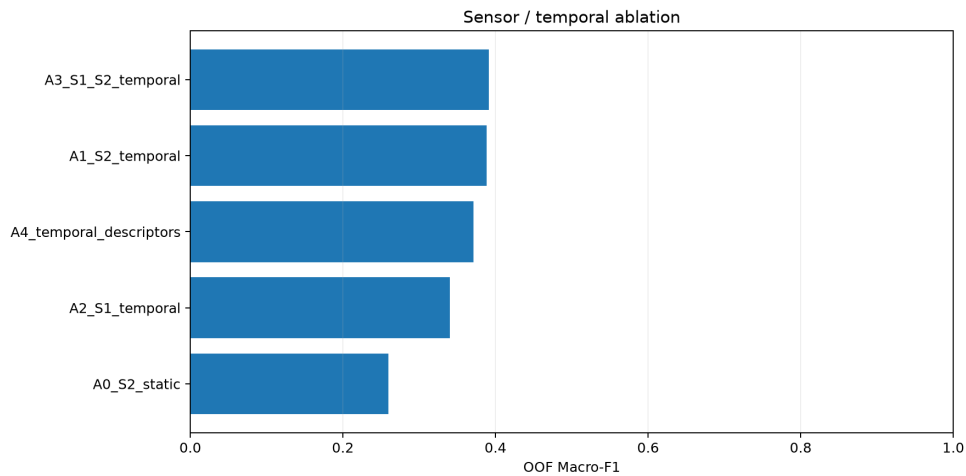


Figure 2. AgriFieldNet family leaders; selection uses the reported OOF results.

The deep sequence results are lower in this run: CNN1D reaches macro-F1 0.343, GRU 0.318 and LSTM 0.319. The fusion leader obtains F1 0.825 for Wheat but zero for Coriander and Green pea; Lentil is 0.015, Potato 0.078 and Rice 0.132. These failures explain why 73.30% accuracy should not be read as reliable recognition of every crop.

V. RESULTS (CONTINUED)

C. Cloud-Gap Robustness

AgriFieldNet scenario	S2-only macro-F1	Fusion macro-F1
No added masking	0.389	0.391
20% random masking	0.363 +/- 0.006	0.367 +/- 0.010
40% random masking	0.306 +/- 0.008	0.327 +/- 0.008
60% random masking	0.194 +/- 0.004	0.259 +/- 0.007
20-day consecutive gap	0.375 +/- 0.009	0.377 +/- 0.006
40-day consecutive gap	0.350 +/- 0.008	0.362 +/- 0.007

TABLE IV. Mean +/- sample SD across fifteen added-mask repeats; baseline n = 1.

The selected AgriFieldNet fusion pipeline degrades less under severe random masking. At 60%, its absolute macro-F1 loss is about 0.132, compared with 0.194 for optical-only. Both pipelines still lose substantial performance. The experiment removes signal columns while retaining optical acquisition diagnostics, and should be interpreted within that design.

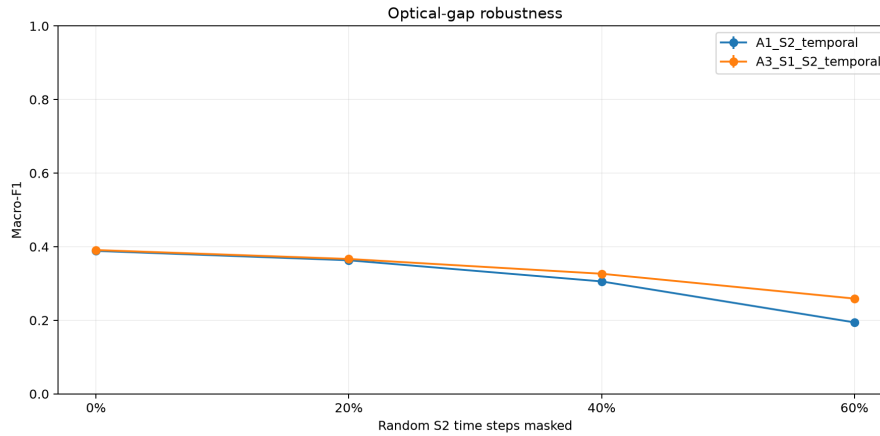


Figure 3. AgriFieldNet sensitivity to additional missing optical signal periods.

Haridwar shows a similar direction in its separate experiment: mean macro-F1 at 60% additional masking is about 0.424 for fusion and 0.267 for optical-only, from baselines of 0.493 and 0.461. Natural optical usability is 60.5% and 73.3% across the two seasons. The Haridwar and AgriFieldNet scores cannot be pooled because their targets, sample units and validation schemes differ.

D. Karnal Natural Missingness and Application Integration

Karnal optical coverage is 100% on 1 June 2025, 31.18% on 17 June, 1.62% on 3 July and zero on 19 July and 4 August. It recovers to 25.68% on 20 August, approximately 100% on the September and early-October dates, and 41.89% on 23 October. Sentinel-1 is valid at every location on these ten dates. Carried application values are not counted as fresh observations in this audit.

The complete application links Karnal monitoring and persistent inference to the two static research pages. The AgriFieldNet builder reconciles metrics against 460,733 OOF predictions across 83 configurations and publishes per-crop results, confusion counts, coverage, robustness, exploratory importance and source hashes. This is evidence publication, not deployment of the AgriFieldNet classifier into the Karnal advisory service.

VI. DISCUSSION AND COMPARISON WITH PRIOR WORK

The results support a practical role for temporal and multisensor information, but their strength depends on the comparison. AgriFieldNet temporal optical features perform substantially better than the static optical family in this run. Fusion and temporal optical leaders are nearly tied at baseline, while the selected fusion pipeline retains more macro-F1 under severe optical masking. The clearest distinction is therefore missing-signal resilience within this experiment, rather than a large baseline accuracy gain.

The Haridwar work extends the published chronology into a five-class radar-optical problem. Kumar and Khare [8] studied phenological dates using field observations, PhenoCam and satellite data. Their day-based errors cannot be ranked against the present classification macro-F1. Similarly, the original AgriFieldNet competition [10,11] uses a different input and evaluation setting. Neither challenge rankings nor published multisensor accuracies [4-6] are interchangeable with these spatial OOF estimates.

Application integration contributes to interpretation. Karnal shows when an optical value is carried, Haridwar exposes the anchors behind a stage label, and AgriFieldNet shows the crops hidden by an aggregate score. Five Karnal personas adapt the interface, while research pages provide the more detailed comparisons. This makes the evidence easier to inspect without implying that interface roles provide production authorization.

VII. NOVELTY BOUNDARY, LIMITATIONS AND VALIDITY

The contribution is the additional temporal processing, explicit supervision boundaries, completed evaluation and application integration. Sentinel fusion, Haridwar phenology and AgriFieldNet classification each have prior work. Dataset creation remains credited to its original authors and organizations.

Haridwar covers one site and two seasons, and the stage labels between anchors are derived. The terminal class has only one row. AgriFieldNet is imbalanced, with fourteen Coriander fields compared with 2,031 Wheat fields, and the selected fusion model fails on several rare classes. Model leaders are chosen from the same OOF evaluations used for reporting; no nested selection estimate, untouched final test or significance claim is provided.

Spatial grid groups reduce one form of overlap but do not test all sources of geographic dependence. The temporal feature window must remain tied to the archived extraction metadata. Natural radar availability is not complete in all AgriFieldNet periods, even though it is complete in the Karnal table. Sensor resampling also does not increase native spatial resolution.

The optical-gap tests are controlled perturbations, not physical cloud simulations. Repeat SD is not a confidence interval across farms or regions. Quick permutation importance is sensitive to sample size and correlated predictors; the dashboard's signed individual-feature ranking differs from the saved plot's positive base-feature aggregation.

Karnal pseudo-label scores remain implementation evidence. Its simplified water rules have not been validated against irrigation events, soil-moisture observations or yield outcomes. The research pages add no Haridwar irrigation advice and no live AgriFieldNet inference. These boundaries prevent completion of the software workflow from being mistaken for operational agronomic validation.

VIII. CONCLUSION AND FUTURE WORK

AgriSense now connects a Karnal operational prototype, a credited Haridwar phenology reanalysis and a completed AgriFieldNet temporal benchmark. Haridwar fusion reaches macro-F1 0.493. AgriFieldNet fusion reaches 0.391 and 73.30% accuracy, with a small baseline difference from temporal optical input and better retention under the tested severe optical masking. Rare-class failures remain a central limitation.

Future work should collect independent Karnal labels and irrigation outcomes, extend Haridwar coverage, and reserve a final holdout after model selection. Matched-estimator ablations, realistic joint signal-and-quality masking, rare-class analysis and larger importance budgets would strengthen the research evidence. A frozen release should preserve extraction dates, manifests, hashes and exact figures.

ACKNOWLEDGEMENT AND ATTRIBUTION STATEMENT

We acknowledge Akash Kumar and Siddhartha Khare for the Haridwar field observations and published chronology, and preserve credit to KVK Dhanauri and the field support acknowledged in their source study [8,9]. Radiant Earth Foundation and IDinsight are credited for AgriFieldNet; IDinsight's Data on Demand team collected its reference labels [10]. We acknowledge the prior challenge work of Tuo, Emelike and Ogundare [11], the Copernicus programme and Google Earth Engine [1-3]. The present authors did not collect the original Haridwar or AgriFieldNet field observations.

DATA AND CODE AVAILABILITY

Haridwar resources are available through the source study and Figshare record [8,9]. AgriFieldNet is identified by DOI 10.34911/rdnt.wu92p1 [10]. AgriSense code and derived workflows are maintained in the project repository [12]. The reviewed local output set contains the completed benchmark, source CSVs and dashboard snapshot; a publication-specific immutable archive remains to be prepared.

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